



Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: spending, segments, preferences, subscriptions



Dataset Summary

Size

Rows: 3,900 • Columns: 18

Key Features

Demographics, Purchase details,
Behavior

Missing Data

37 missing values in Review Rating

EDA & Preparation (Python)

01

Load & Inspect

pandas: df.info(), describe()

02

Clean & Impute

Imputed Review Rating by category median

03

Standardize

Renamed columns to snake_case

04

Feature Eng.

age_group, purchase_frequency_days

05

DB

Loaded cleaned DataFrame into MySQL

Descriptive Stats Snapshot

Summary	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
✓ Metrics	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
✓ mean	1915750000	44.636462	59.764339	3.450683	23.531538
✓ std	1175378738	15.307876	23.885342	0.716783	14.478195
✓ min	10000000	18.000000	20.000000	2.200000	1.000000
✓ 25%	975750000	31.000000	39.000000	3.100000	13.000000
✓ 50%	1945250000	46.000000	60.000000	3.800000	23.000000
✓ 75%	2925290000	57.000000	81.000000	4.400000	33.000000
✓ max	3940300000	70.000000	100.000000	5.000000	50.000000
✓ 25% ↓ 49.05%	975750000 ↓ 49.05%	31.000000 ↓ 30.60%	39.000000 ↓ 34.82%	3.100000 ↓ 10.17%	13.000000 ↓ 44.76%
✓ 50% ↓ 1.63%	1945250000 ↑ 1.63%	46.000000 ↑ 3.05%	60.000000 ↑ 0.40%	3.800000 ↑ 10.11%	23.000000 ↓ 2.26%
✓ 75% ↑ 52.75%	2925290000 ↑ 52.75%	57.000000 ↑ 27.74%	81.000000 ↑ 35.43%	4.400000 ↑ 27.50%	33.000000 ↑ 40.25%
✓ max ↑ 105.71%	3940300000 ↑ 105.71%	70.000000 ↑ 57.07%	100.000000 ↑ 67.24%	5.000000 ↑ 44.83%	50.000000 ↑ 112.49%

Highlights

- Mean purchase: \$59.76
- Avg review rating: 3.75 (37 missing)
- Top item frequency: Blouse (171)

SQL Findings — Revenue & Segments



Revenue by Gender

Male: 157,890 • Female: 75,191



Customer Segments

Loyal: 3,116 • Returning: 701 • New: 83



Repeat Buyers & Subscriptions

Repeat buyers more likely to subscribe
(958 vs 2,518)

Top Products & Ratings



Gloves

Avg rating: 3.86



Sandals

Avg rating: 3.84



Boots

Avg rating: 3.82



Hat

Avg rating: 3.80



Skirt

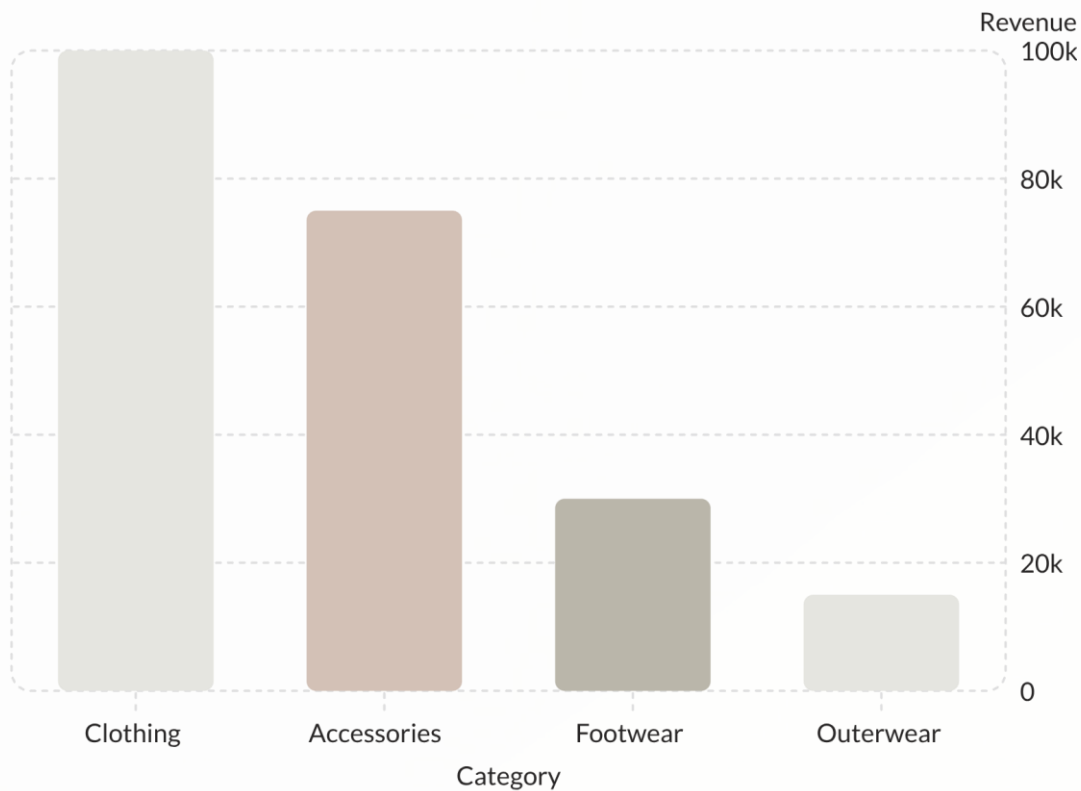
Avg rating: 3.78

The illustration shows several shopping tags and receipts. The tags are colorful and feature various text and symbols. One tag says 'MAP PAL OsLE', another says 'SHOPPING SALE', and a third says 'SHOPOG AI SFORMETTUCN'. There are also receipts with a total of \$4563.00 and another receipt with a total of \$900. The tags and receipts are arranged in a scattered manner, with some overlapping.

Top: Hat 50% • Sneakers 49.66% •
Coat 49.07% • Sweater 48.17% •
Pants 47.37%

Examples: customer_id 3 (\$73), 4 (\$90), 29 (\$94) — spent above average despite discounts

Revenue & Shipping Insights



Shipping Comparison

Express avg: \$60 • Standard avg: \$58

Age Group Revenue

Young Adult: 62,143 • Middle-aged: 59,197 • Adult: 55,978 • Senior: 55,763

Dashboard Preview



Key Metrics

Avg purchase \$59.76 • Avg rating 3.75 • Customers 3.9K

Subscription Mix

Yes 27% • No 73%

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Discount Policy

Balance sales lift with margin control

Product Positioning

Feature top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users